

Breakout Volatility Expansion Quantitative Strategy Engine

BVEQSE v1.0

Product Requirements Document

Research, Backtesting, Validation and Forward-Observation System

Document status:	Draft for implementation (specification freeze pending)
Document date:	29 September 2026
Strategy family:	Long-only Indian equities, daily bars
Sibling strategy:	TPQSE v1.0 (read-only; MUST NOT be modified)
Target implementer:	AI coding agent / engineering team
Claim status:	No edge is assumed. No profitability is claimed.

This document is a research specification. All numeric strategy values are baseline research starting points, not optimized or recommended parameters. Regulatory fee values are placeholders that must be verified against current exchange and broker schedules when the cost configuration is created.

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Executive Summary

BVEQSE v1.0 is a second, independent, long-only systematic strategy for liquid Indian equities. It tests whether a *breakout after consolidation and volatility compression, confirmed by abnormal volume* carries continuation information that survives realistic costs, slippage, out-of-sample and walk-forward validation. It is deliberately structurally different from TPQSE (EMA(50) trend, RSI(14) pullback recovery, ATR stop, 2R target).

This PRD specifies (i) a fully deterministic baseline rule set, (ii) a bounded research grid with a pre-registered selection protocol, (iii) an event-driven backtester with conservative execution semantics, (iv) statistical validation (cluster bootstrap, Monte Carlo, random-control tests, deflated Sharpe), (v) expanding-window walk-forward validation with purge rules, (vi) comparison against TPQSE, (vii) an independent forward-observation engine with immutable versioning, and (viii) objective acceptance criteria and failure flags.

Key design decisions and ambiguities resolved in this document (full register in Appendix A.1):

- **Compression timing.** Comparing $ATR_{14,t}$ (which contains the breakout candle’s expanded true range) against a prior median is nearly self-contradictory for a breakout signal. The baseline measures compression on the *pre-breakout state* at $t - 1$, using price-normalized ATR. The stop still uses $ATR_{14,t}$ frozen at signal close, as specified.
- **Stop/entry circularity.** $Stop = Entry - k \cdot ATR$ needs an entry price that is unknown at signal time. Resolution: the stop and target are computed at fill time from the actual fill price and the frozen signal-time ATR; $R = k \cdot ATR_{signal}$ exactly.
- **Unspecified portfolio constraints** (maximum concurrent positions, position weight cap, liquidity floor, signal ranking, time stop) are made explicit, configurable, and frozen before any test.
- **“Negative ending equity” in Monte Carlo** cannot occur under fractional-risk compounding; it is replaced by explicit loss/ruin probabilities.
- **Survivorship** is marked UNRESOLVED for the initial universe and caps the maximum achievable status (Section 25).

1 Background

TPQSE v1.0 is a completed long-only trend-pullback system with a rigorous backtesting and forward paper-observation framework. It contains an acknowledged, unresolved survivorship bias because its historical universe consists of currently active NSE stocks. BVEQSE is built as a separate system so that (a) it can be audited without contaminating TPQSE, (b) trade behaviour can be compared, and (c) a combined portfolio can be evaluated *only after* each strategy has independently completed research.

Independence rules. BVEQSE code, configuration, state stores, and result directories **MUST NOT** import from, write to, or modify TPQSE artifacts. TPQSE outputs (trade logs, equity curves) are consumed strictly read-only, and only by the comparison module (Section 17).

2 Problem Statement

Primary research question. Can a systematic breakout strategy identify statistically meaningful continuation moves in liquid Indian equities after price consolidation and volatility compression, while remaining robust after realistic transaction costs, slippage, out-of-sample testing, and walk-forward validation?

Why a PRD is needed. Breakout strategies are highly parameter-sensitive, prone to look-ahead errors (resistance defined by the breakout candle itself), execution optimism (gap entries, intrabar ambiguity), and multiple-testing inflation. Without a precise specification, an implementing agent would silently make dozens of consequential assumptions. This document removes those degrees of freedom.

Non-goals for v1.0. No machine learning, no short selling, no leverage, no intraday signal generation, no pyramiding or averaging down, no derivatives, no automated live order routing, no automatic parameter adaptation.

Pre-design analysis (summary). The twelve-point analysis required before authoring is delivered in Appendix A.1 (ambiguities), A.2 (look-ahead sources), A.3 (survivorship), A.4 (backtesting errors), A.5 (statistical pitfalls), A.6 (external dependencies), and A.7 (configurable versus frozen items).

3 Research Objectives

3.1 Primary objective

Determine whether the BVEQSE hypothesis survives testing. A negative result is a valid, publishable-internally outcome and **MUST** be recorded as “Failed”.

3.2 Secondary questions

ID	Question	Primary instrument
Q1	Does volatility compression before a breakout contain useful information?	Ablation (Exp. G, ABL)
Q2	Does abnormal volume improve breakout quality?	Ablation (Exp. H, ABL)
Q3	Does trend direction improve breakout reliability?	Ablation (Exp. I, ABL)
Q4	Sensitivity to breakout lookback?	Exp. C, plateau analysis
Q5	Sensitivity to ATR stop distance?	Exp. E
Q6	Sensitivity to reward/risk target?	Exp. F
Q7	Behaviour across market regimes?	Regime analysis (Sec. 16)
Q8	Sufficiently different from TPQSE?	Comparison (Sec. 17)
Q9	Survives Indian execution costs?	Cost sensitivity 1x/2x/3x
Q10	Edge remains after strict OOS and walk-forward?	Walk-forward (Sec. 13)

3.3 Statistical hypotheses

Let μ_R denote the population mean of net R-multiple per trade under the specified execution model.

$$H_0 : \mu_R \leq 0 \quad \text{versus} \quad H_1 : \mu_R > 0.$$

Rejecting H_0 on historical data is necessary but never sufficient for “Validated” status.

4 Strategy Hypothesis

Mechanistic conjecture (untested). Periods of narrow price range and contracting volatility reflect balance between buyers and sellers. A close above prior resistance on abnormal volume may indicate new participation and short-term continuation. This is a conjecture motivating the rule set; it **MUST NOT** be presented as an established market effect.

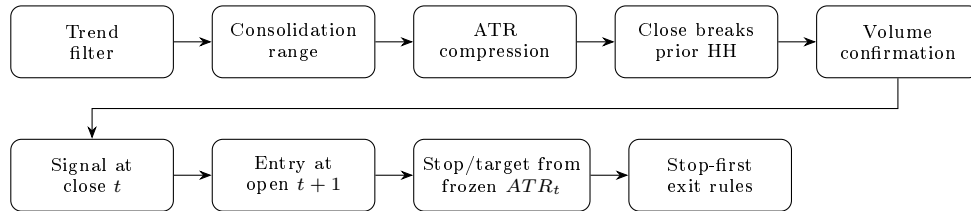


Figure 1: Conceptual signal pipeline. Each stage is an independently ablatable filter except the breakout condition.

Distinctness from TPQSE. TPQSE buys *pullback recovery within a trend*; BVEQSE buys *range expansion after compression*. Both are long-only trend-adjacent systems and may be positively correlated through market beta; the degree is an empirical question (Section 17).

5 Strategy Specification

5.1 Notation

Sessions are indexed by the exchange trading calendar (NSE). t is the signal session (breakout candle); $t + 1$ is the next eligible session. O, H, L, C, V are open, high, low, close, volume of the (split/bonus-adjusted) daily bar. All quantities at index t are computed *only* from bars with index $\leq t$ and, where subscript $t - 1$ appears, only from bars $\leq t - 1$.

5.2 Indicator definitions (deterministic)

EMA. With $\alpha = 2/(N_E + 1)$: $E_{N_E} = \frac{1}{N_E} \sum_{i=1}^{N_E} C_i$ (SMA seed on the first N_E valid bars), and $E_t = \alpha C_t + (1 - \alpha)E_{t-1}$ for $t > N_E$. Library defaults with different seeding **MUST NOT** be used unless proven numerically identical.

True range and ATR (Wilder).

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|), \quad t \geq 2,$$

$$ATR_{N_A, N_A+1} = \frac{1}{N_A} \sum_{i=2}^{N_A+1} TR_i, \quad ATR_t = \frac{(N_A - 1) ATR_{t-1} + TR_t}{N_A}.$$

Normalized ATR. $NATR_t = ATR_{N_A, t}/C_t$ (dimensionless; comparable across price levels and across time as price drifts).

Prior-window extremes and range.

$$HH_{L, t-1} = \max_{i=t-L}^{t-1} H_i, \quad LL_{L, t-1} = \min_{i=t-L}^{t-1} L_i, \quad Range_{L, t-1} = \frac{HH_{L, t-1} - LL_{L, t-1}}{LL_{L, t-1}}.$$

Volume average. $\bar{V}_{L_v, t-1} = \frac{1}{L_v} \sum_{i=t-L_v}^{t-1} V_i$ (simple mean, prior completed sessions only; zero-volume days count as zero; any missing value invalidates the window).

5.3 Baseline conditions

ID	Condition (true at close of t)	Rationale and notes
C1	$C_t > E_t$	Price above medium-term trend. Uses close t (known at signal time).
C2	$E_t > E_{t-5}$	Trend slope positive over 5 sessions; avoids stale/flat trend.
C3	$Range_{L, t-1} \leq \rho$, baseline $\rho = 10\%$, $L = 20$	Prior 20 sessions were compressed. Threshold is a research starting point, not optimal.
C4	$NATR_{t-1} < \text{median}\{NATR_{t-1-L_a}, \dots, NATR_{t-2}\}$, $L_a = 20$	Volatility contraction measured on the pre-breakout state; the breakout bar is excluded from both sides. Median-based test passes roughly half of bars by construction, so it is a weak filter; stricter variants are in the grid.
C5	$C_t > HH_{L, t-1}$	Breakout above the highest high of the prior L completed sessions. Resistance excludes bar t .
C6	$V_t > \mu_v \cdot \bar{V}_{L_v, t-1}$, $\mu_v = 1.5$, $L_v = 20$; and $V_t > 0$	Abnormal participation, denominator strictly prior.
C7	$\text{median}_{i=t-60}^{t-1}(C_i V_i) \geq \Lambda$ (baseline $\Lambda =$ INR 10 crore)	Liquidity floor (assumed by “liquid Indian equities”; not in the original brief; frozen and ablated in X4).
C8	Bar t valid, data-quality gates passed (Sec. 9), warm-up satisfied, no open position or pending entry in symbol	Eligibility.

Signal: $Signal_t = C1 \wedge C2 \wedge C3 \wedge C4 \wedge C5 \wedge C6 \wedge C7 \wedge C8$.

5.4 Warm-up (explicit)

No signal at t unless all inputs exist without imputation. For parameter vector θ , the first eligible bar index (1-based, counting valid consecutive calendar sessions) is

$$W(\theta) = \max(N_E + 5, N_A + L_a + 2, L + 1, L_v + 1, 62).$$

Baseline: $W = \max(55, 36, 21, 21, 62) = 62$ (the 62 arises from the 60-session liquidity window plus current bar plus one). Research data are loaded from **2017-01-01** so that the first trading date of the research window (2018-01-01) is fully warmed up.

5.5 Parameter registry: baseline versus research grid

ID	Parameter	Baseline	Research grid (bounded)	Status
P01	N_E EMA period	50	fixed	Frozen
P02	Slope lag	5	fixed	Frozen
P03	L (breakout/consolidation lookback, $L_b = L_c$)	20	10, 15, 20, 30, 40	Researched
P04	ρ range threshold	10%	5, 7.5, 10, 12.5, 15 (%)	Researched
P05	Compression rule	median, $L_a = 20$	off; median; 25th percentile	Ablation
P06	N_A ATR period	14	fixed	Frozen
P07	μ_v volume multiple	1.5	off; 1.25; 1.5; 2.0; 2.5	Researched
P08	L_v	20	fixed	Frozen
P09	k stop multiple	1.5	1.0, 1.25, 1.5, 1.75, 2.0	Researched
P10	m target (R)	2.0	1.0, 1.5, 2.0, 2.5, 3.0, 4.0	Researched
P11	Risk fraction f	0.5%	0.25, 0.5, 1.0 (%) sizing sensitivity only	Never optimized
P12	Slippage per leg	0.05%	via cost multipliers 1x, 2x, 3x	Sensitivity
P13	Max concurrent positions	10	5, 10, 20	X1
P14	Max position weight	20% of equity	10, 20, 33%	X1
P15	Liquidity floor Λ	INR 10 cr	5, 10, 25	X4
P16	Max ADV participation	2% of median 20-day volume	1, 2, 5%	X4
P17	Max holding (time stop)	none	none; 40; 60; 80 sessions	X2
P18	Gap-up entry filter	disabled	disabled; skip if gap > 3%; > 5%	X3
P19	Signal ranking (capacity conflicts)	volume ratio desc., symbol asc. tie-break	volume ratio; range asc.; alphabetical	X1
P20	Price series	split/bonus-adjusted, not dividend-adjusted	fully adjusted (Adj Close basis)	X5

Only P03, P04, P07, P09, P10 enter walk-forward parameter selection (Section 13). All others are stress or ablation dimensions evaluated around the frozen baseline; they **MUST NOT** be selected using test-period data.

6 Signal Generation

6.1 Timing and confirmation

Signals are generated only after bar t has closed, on a completed daily bar. Intrabar signal generation is prohibited. The signal timestamp is t_{close} (15:30 IST on session t); the implementation **MUST** store both the bar date and the wall-clock time at which the signal was computed (forward mode).

6.2 Look-ahead prohibitions

The implementation **MUST NOT** use: future candles; the final high of bar t in its own resistance; indicators recomputed with data beyond t ; future volume; future corporate-action or universe information; full-sample normalization or percentile ranks; and current-day market-cap or sector classification as if historical (unless explicitly labelled).

Mandatory structural safeguards.

1. Indicator functions take an “as-of” index and slice history to $\leq$ as-of before computing.
2. **Truncation invariance:** signals computed on data truncated at t **MUST** equal signals for t computed on the full history (automated test T-LA1).
3. **Future-scramble invariance:** replacing all bars after t with random values **MUST NOT** change any signal at or before t (T-LA2).

6.3 Signal record

Every evaluated candidate that satisfies C1–C7 (regardless of later rejection) **MUST** be persisted with a status from: ACCEPTED, SUPPRESSED_OPEN_POSITION, REJECTED_CAPACITY, REJECTED_CASH, REJECTED_LIQUIDITY_CAP, REJECTED_GAP_FILTER, EXPIRED_NO_DATA, EXPIRED_INVALID_BAR, EXPIRED_CIRCUIT_LOCK. Signals **MUST NOT** be deleted; this prevents silently removing losing (or winning) setups.

6.4 Duplicate and overlapping signals

`signal_id` = SHA-256 of (strategy version || symbol || signal date || config hash), truncated to 16 hex characters. Re-running any session **MUST** be idempotent: an existing `signal_id` is never regenerated or altered. A signal in a symbol that already has an open position or pending entry is suppressed and logged. A symbol whose position exits intraday on day d may generate a new signal at the close of d .

7 Execution Model

7.1 Entry

Entry is at the **open of the next eligible trading session** $t + 1$ (per NSE calendar):

$$F^{entry} = Open_{t+1} \times (1 + s_{buy}), \quad s_{buy} = 0.0005 \text{ (baseline).}$$

Case	Required behaviour
Normal entry	Fill at F^{entry} ; stop and target computed from F^{entry} (Sec. 8).
Gap-up entry	Fill at the gapped open (plus slippage). No price cap in baseline. Record $gap_pct = Open_{t+1}/C_t - 1$. Optional filter P18 (research only).
Gap-down entry	Fill at the open. Since $stop = F^{entry} - kATR$ is always below the fill, the stop is not breached at the fill.
Missing next-session data	Session $t + 1$ is a calendar trading day but the symbol has no bar: signal expires (EXPIRED_NO_DATA). No delayed entry, no stale price, no forward-fill.
Market holiday	“Next eligible session” is the next date on the NSE trading calendar; weekends/holidays are skipped by calendar, not by data presence.
Stale data	Bar identical to the prior bar in OHLC with zero volume, or flagged stale by validation: treated as invalid; signal expires (EXPIRED_INVALID_BAR).
Invalid OHLC	Any of $H < \max(O, C, L)$, $L > \min(O, C, H)$, non-positive price, negative volume, NaN: bar invalid; expiry as above.
Circuit-lock heuristic	If $O = H = L = C$ on $t + 1$ with $Open_{t+1} > C_t$, the buy is assumed unfillable: EXPIRED_CIRCUIT_LOCK. Configurable; default on. Limitation: daily bars cannot prove lock state.
Duplicate signal	Ignored via <code>signal_id</code> uniqueness.

Case	Required behaviour
Already-open position	Suppressed (one position per symbol).
Structural rules	No pyramiding, no averaging down, no leverage, no short selling, whole shares only.

7.2 Exit evaluation order

Stop and target become active from the entry bar $t + 1$ (using that bar's H and L , which occur after the open fill). For each open position on each session d , evaluated in this order:

1. **Gap through stop:** if $Open_d \leq S$: exit at $Open_d \times (1 - s_{sell})$ (not at S).
2. **Gap through target:** else if $Open_d \geq T$: exit at $Open_d \times (1 - s_{sell})$ (not at T).
3. **Stop touched:** else if $Low_d \leq S$: exit at $S \times (1 - s_{sell})$ (stop-market with slippage).
4. **Target touched:** else if $High_d \geq T$: exit at $T \times (1 - s_{sell})$ when `apply_slippage_on_limit_exits=true` (baseline, conservative and consistent).

Intrabar ambiguity: if one bar touches both S and T (no gap), **STOP FIRST**. Lower-circuit lock heuristic: if $O = H = L = C$ and $C < C_{prev}$ on an exit day, the exit is deferred to the next unlocked session's open (flagged `EXIT_DEFERRED_LOCK`).

End of data / permanent data termination. Positions open at the dataset end are *censored*: marked at last close, flagged `CENSORED`, and excluded from closed-trade statistics but included in equity. If a symbol's data terminates permanently while held, exit at last available close, flag `DATA_TERMINATION`, and report a sensitivity treating it as -100% of position value. No returns are fabricated.

7.3 Daily event-loop order (backtest and forward, identical semantics)

For each trading session d in chronological order:

1. **Snapshot** $Eq_{d-1} = Cash + \sum q_i C_{i,d-1}$ (close marks of session $d - 1$).
2. **Open-gap exits:** process rules 1–2 for carried positions; credit cash after costs.
3. **Pending entries:** rank pending signals per P19; for each, validate bar d , compute fill, stop, target, quantity from Eq_{d-1} ; apply caps; deduct cash and costs.
4. **Intraday exits:** rules 3–4 for all open positions including those entered today.
5. **Mark to market** at C_d ; record equity, exposure.
6. **Signal generation** at close of d for all eligible symbols; persist signals; queue accepted pending entries for $d + 1$.

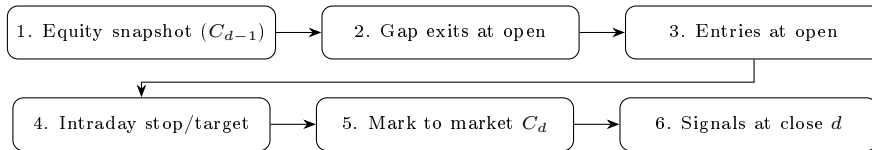


Figure 2: Per-session event order. Sizing equity is frozen at step 1.

7.4 Slippage semantics

Slippage worsens the *fill price* (buy up, sell down). It is never applied as an arbitrary P&L subtraction. Baseline $s_{buy} = s_{sell} = 0.05\%$ (about 0.10% round trip). Backtest and forward engines **MUST** call the same execution function. Tick-size rounding (NSE 0.05 INR, buys rounded up and sells rounded down) is supported but disabled by default because back-adjusted prices are not tick-aligned.

7.5 Transaction costs

A central `costs.py` configuration (versioned, with `effective_date` and `source_note`) **MUST** hold all rates; strategy logic **MUST NOT** contain fee constants. Costs are computed per leg on turnover $= F \times q$ using *fill* prices and are deducted from cash at fill time.

Component	Baseline placeholder	Application
Brokerage	0 (delivery, discount broker) or config % with cap	Per leg; configurable percentage, flat fee, cap.
STT	0.1% buy and sell (delivery)	On turnover, both legs.
Exchange transaction charge	NSE cash-segment rate (verify)	On turnover, both legs.
SEBI turnover fee	INR 10 per crore (verify)	On turnover, both legs.
GST	18%	On brokerage + exchange charge + SEBI fee (not on STT or stamp duty).
Stamp duty	0.015% (verify)	Buy leg only.
DP charge	flat per scrip per sell day (optional, verify)	Sell leg, plus GST.
Slippage	0.05% per leg	In fill price (see above); not a fee.

All regulatory values above are placeholders as understood at authoring time and **MUST** be verified against current NSE/SEBI/broker schedules when the config is first created; the verification date and source are stored in the config and hashed. Cost sensitivity multiplies all fee components and slippage by $1\times, 2\times, 3\times$.

8 Risk Management

8.1 Frozen stop distance

$$R = k \cdot ATR_{N_A, t}, \quad S = F^{entry} - R, \quad T = F^{entry} + m R.$$

$ATR_{N_A, t}$ is captured at signal close and stored as `atr_signal`. It **MUST NOT** be recalculated after signal creation; stop and target are immutable once the position is opened (no trailing, no dynamic ATR). An automated test asserts S and T are constant over the holding period.

8.2 Position sizing

$$Equity = Cash + MarketValue, \quad RiskAmount = f \cdot Eq_{d-1}, \quad f = 0.005,$$

$$q_{risk} = \left\lfloor \frac{RiskAmount}{R} \right\rfloor.$$

Eq_{d-1} uses mark-to-market equity (cash plus open positions valued at the last completed close), *never* free cash alone. Final quantity:

$$q = \min \left(q_{risk}, \left\lfloor \frac{w_{max} Eq_{d-1}}{F^{entry}} \right\rfloor, \left\lfloor \frac{p_{adv} \cdot \overline{V}_{20}^{med}}{1} \right\rfloor, \left\lfloor \frac{Cash_{avail}}{F^{entry}(1+\kappa)} \right\rfloor \right),$$

where κ is a conservative fee buffer. If $q < 1$ the signal is rejected with an explicit reason. Because ATR stops are commonly 2–6% of price, uncapped risk sizing can imply large position weights; caps P13–P16 are therefore mandatory and frozen.

8.3 Trade-level R definitions

Planned risk = $q \cdot R$. $R_{price} = (F^{exit} - F^{entry})/R$ (after slippage, before fees). $R_{net} = (q(F^{exit} - F^{entry}) - Fees)/(qR)$. A *frictionless* R_{fl} (no slippage, no fees) is also stored for diagnostics. **Expectancy is defined on R_{net} unless stated.**

8.4 Portfolio-level controls

No leverage: cash cannot go negative. Maximum concurrent positions and weights per P13–P14. There is no automated portfolio-level stop; drawdown thresholds only *flag* (Section 19).

9 Data Architecture

9.1 Strategy input versus research-only context

Category	Contents and rules
Strategy input data	Per symbol: date, open, high, low, close, volume, symbol. Universe eligibility as of date. Only these (and derived indicators) may enter signal generation.
Research-only context	Adjusted-close variants, corporate-action tables, index data (regime), sector, market cap, TPQSE outputs. MUST NOT feed signal generation, sizing, or exits. Regime, sector and cap are attached to trades <i>after</i> the fact for analysis (and stored as ML metadata).

9.2 Baseline price series

Source: yfinance (auto_adjust=False), NSE tickers (suffix .NS), daily. Baseline uses the split/bonus-adjusted OHLC fields and split-adjusted volume; dividend adjustment is *not* applied (P20). Fully adjusted series are a sensitivity run (X5).

9.3 Validation gates (data/validation.py)

1. Schema, dtype, timezone (IST dates), strictly increasing unique dates.
2. OHLC invariants; positive prices; volume ≥ 0 .
3. Calendar completeness: every NSE trading day between first and last bar present; missing sessions listed. Symbols with coverage below 95% are excluded *by a rule fixed before any performance is computed*, and listed.
4. Missing values are never imputed or forward-filled. A missing bar breaks indicator chains: rolling windows that include a missing session are undefined; EMA/ATR recursions reset and must re-seed on the next consecutive valid bars.
5. Stale-bar detection (repeated identical OHLC with zero volume).
6. Suspected corporate-action discontinuity: $|C_t/C_{t-1} - 1| > 35\%$ flags CA_SUSPECT for review. Affected trades are reported with and without the flag; exclusion is permitted only by this pre-registered rule applied symmetrically to winners and losers.
7. Zero and missing volume: zero volume is a real observation (fails C6 at t); NaN volume in the required window invalidates the signal.

Each validation run outputs a machine-readable data-quality report stored with the experiment.

9.4 Corporate actions and Yahoo/yfinance limitations

Event	Handling and limitation
Splits, bonuses	Relied on vendor back-adjustment; inconsistencies possible for Indian bonus issues; detected only heuristically (CA_SUSPECT).
Dividends	Not in baseline prices; dividend income is <i>omitted</i> (understates long-holding returns; distorts stops around ex-dates). Sensitivity X5.
Rights issues	Frequently unadjusted or inconsistently adjusted; unresolved limitation.
Mergers, acquisitions	Vendor continuity uncertain; discontinuities flagged; no synthetic stitching.
Ticker changes	Explicit <code>symbol_map</code> table with effective dates. Silent substitution is prohibited; unmapped symbols are excluded and logged.
Delistings	Generally absent from yfinance; see Section 15.

Implications of back-adjusted prices. Historical adjusted prices are not the prices at which historical trades were executable. Share quantities, tick alignment, lot sizes, absolute INR liquidity, and stop levels in INR are

approximate; percentage-based logic is more reliable than absolute levels. Adjusted prices **MUST NOT** be described as perfectly reproducing historical executable prices. Vendor revisions can change history; each data pull stores a content hash and timestamp.

10 Universe Definition

Initial universe: `CURRENT_ACTIVE_UNIVERSE`, reusing existing universe infrastructure where possible (read-only import of the symbol list; no writing into TPQSE stores).

Interface requirement. The engine consumes a `UniverseProvider` with method semantics $Universe_t = \text{members}(\text{as_of}=t)$. The current-active provider returns the same list for every t (flagged `point_in_time=false`); a future point-in-time provider can be substituted without engine changes.

Experiment immutability. Every experiment records: universe version, size, creation timestamp, and a snapshot file (symbol list plus SHA-256). Membership **MUST NOT** change silently between experiments; a new list means a new universe version and rerun.

Exclusions. Symbols failing coverage, liquidity (C7), or mapping rules are excluded with logged reasons; exclusion rules use only information available at each date (liquidity is evaluated point-in-time per date).

11 Backtesting Framework

11.1 Design

Indicators and candidate signals may be computed in a vectorized per-symbol pass using strictly backward-looking operations; portfolio simulation **MUST** be event-driven per Section 7. Both layers are verified for equivalence against a slow reference implementation on synthetic and sampled real data.

11.2 Initial conditions

Starting equity INR 1,000,000 (configurable); all cash; research window 2018-01-01 to research-freeze date; reserved final holdout H: 2026-01-01 to the freeze date, evaluated exactly once after all research decisions are locked.

11.3 Experiments

ID	Purpose	Variation (all other parameters at baseline)
A	Baseline	Baseline P01–P20.
B	Consolidation threshold	$\rho \in \{5, 7.5, 10, 12.5, 15\}\%$.
C	Breakout lookback	$L \in \{10, 15, 20, 30, 40\}$ (consolidation window tied to L).
D	Volume multiple	$\mu_v \in \{1.25, 1.5, 2.0, 2.5\}$ and off.
E	ATR stop	$k \in \{1.0, 1.25, 1.5, 1.75, 2.0\}$.
F	Reward/risk	$m \in \{1, 1.5, 2, 2.5, 3, 4\}$.
G	Compression on/off	C4 off versus baseline versus 25th-percentile variant.
H	Volume on/off	C6 off versus baseline.
I	Trend filter on/off	C1+C2 off versus baseline.
ABL	Full ablation matrix	$2^4 = 16$ cells over {trend, consolidation range C3, ATR compression C4, volume C6}; C5 always on.
X1	Portfolio constraints	P13, P14, P19 variants.
X2	Time stop	P17 variants.
X3	Gap-up filter	P18 variants.
X4	Liquidity/capacity	P15, P16 variants.
X5	Price adjustment	Fully adjusted series.
CS	Cost sensitivity	$1\times, 2\times, 3\times$ costs.

Every run increments a persistent `n_trials` counter (total distinct parameter/rule configurations evaluated), used in deflated-Sharpe correction (Section 12).

11.4 Sensitivity and anti-mining protocol

- One-at-a-time sweeps (B–F) around the baseline are descriptive, not selective.
- Plateau criterion: robustness is judged by the fraction of one-step-neighbour grid cells with positive net expectancy, not by the single best cell.
- The best in-sample combination **MUST NOT** be reported as evidence of robustness or promoted as production-ready.
- Grid bounds are frozen before running (Section ??); adding grid values after seeing results requires a new version and counts toward n_{trials} .

12 Statistical Validation

12.1 Required metrics

Level	Metrics and definitions
Trade	Number of trades; win rate ($R_{net} > 0$); loss rate; average R ; median R ; expectancy = $\overline{R_{net}}$; profit factor = $\sum \text{wins} / \sum \text{losses} $ (net INR); average winner, average loser; maximum winner, maximum loser.
Portfolio	Total return; CAGR; annual returns; annualized volatility (daily, 252 days); Sharpe (rf = 0 and rf = configured); maximum drawdown; drawdown duration (peak to trough and peak to recovery); recovery time; exposure (fraction of days with ≥ 1 position, and mean invested value/equity); turnover = annual traded value / mean equity.
Distribution	R histogram and quantiles; win/loss distributions; holding period (sessions); MFE and MAE in R units from daily bars ($MFE = \max(H - F^{entry})/R$, $MAE = \max(F^{entry} - L)/R$ over the hold); monthly and yearly return tables.

12.2 Confidence intervals

Win rate, mean R , and expectancy receive 95% intervals by **10,000-replicate bootstrap**. Because trades entered on nearby dates share market shocks, the *primary* interval is a **cluster bootstrap resampling entry-months**; the iid trade bootstrap is reported for comparison and a Wilson interval is reported for win rate. Seeds derive from the experiment seed (Section 22); replicate count and seed are stored.

12.3 Monte Carlo

At least **10,000** simulations on OOS net- R multiples (or the specified experiment’s trades), with fixed fractional risk f and compounding. Two resampling modes: iid trade resampling and moving-block resampling (block length recorded) to preserve loss clustering. Simulated trade count equals the observed count over the analysed span. Outputs: max drawdown distribution; terminal equity distribution; longest losing streak; return quantiles; $P(Eq_{terminal} < Eq_0)$; $P(MDD > X)$ for $X \in \{20\%, 30\%, 50\%\}$.

“**Negative ending equity**” is not attainable under fractional risk compounding with $R_{net} \geq -1$ -ish losses, so the reported probabilities are terminal loss and threshold-ruin probabilities. **Monte Carlo output is conditional on the historical trade distribution and model assumptions (independence or block structure, no regime change, no concurrency effects); it MUST NOT be presented as a real-world probability.**

12.4 Random-control test

- **Null N1 (random entry)**: each actual trade’s entry date is replaced by a random session drawn uniformly from dates on which the same symbol is eligible (warm-up satisfied, C7 true, bar valid) within the tested period; identical stop/target/cost/sizing logic applied.
- **Null N2 (trend-matched random entry)**: as N1, restricted to dates satisfying C1–C2, isolating the contribution of consolidation, compression, breakout, and volume.
- Statistic: mean R_{net} (secondary: total net return). Simulations: 10,000 per null. Seed: derived from the experiment seed via `numpy.random.default_rng`.
- $p = \frac{1 + \#\{\text{sim} \geq \text{obs}\}}{1 + N_{sim}}$.

- **Limitations:** depends on the null’s definition; random dates in bull markets embed market drift; overlapping trades are not independent; symbol survivorship persists. **Passing does not prove a real-world trading edge.**

12.5 Multiple testing

Report the Deflated Sharpe Ratio using the recorded n_{trials} , skewness and kurtosis of returns, and sample length. A White reality check or SPA-style test across the grid is preferred where practical. Plain significance claims without a defined test are prohibited.

12.6 Cost sensitivity

Run Experiment A (and the walk-forward stitched OOS) at $1\times$, $2\times$, $3\times$ costs and compare CAGR, Sharpe, max drawdown, net P&L, expectancy. A strategy that is profitable only at $1\times$ costs is flagged as cost-fragile.

13 Walk-Forward Validation

13.1 Meaning of “training” for a deterministic strategy

There is no model fitting. “Training” means *parameter selection over the bounded walk-forward grid using only training-period trades*; the selected parameters are then applied unchanged to the test year.

Walk-forward grid (reduced to limit multiple testing): $\rho \in \{7.5, 10, 12.5\}\%$, $L \in \{15, 20, 30\}$, $\mu_v \in \{1.25, 1.5, 2.0\}$, $k \in \{1.25, 1.5, 1.75\}$, $m \in \{1.5, 2.0, 3.0\}$: 243 cells.

Pre-registered selection rule (frozen). For each cell compute training net expectancy $\bar{R}$ over qualifying trades (minimum 100 training trades, else ineligible). Score = mean of $\bar{R}$ across the cell and its one-step grid neighbours (plateau smoothing). Select the highest-scoring eligible cell; ties broken by closeness to baseline then lexicographic order. If no cell is eligible or the best score is ≤ 0 , the fold’s parameters default to the frozen baseline and the fold is flagged NO_POSITIVE_TRAIN_CELL.

13.2 Folds (expanding window)

Fold	Training window	Test year	Notes
1	2018-01-01 – 2020-12-31	2021	Includes the 2020 shock in training.
2	2018-01-01 – 2021-12-31	2022	
3	2018-01-01 – 2022-12-31	2023	
4	2018-01-01 – 2023-12-31	2024	
5	2018-01-01 – 2024-12-31	2025	
H	(frozen after fold 5)	2026-01-01 to freeze	Single-shot holdout; not used for any selection.

13.3 Trade classification and boundary rules

- **Test trades (OOS):** classified by *entry date* within the test year (entry-based OOS).
- **Training trades:** only trades whose *exit date* is on or before the training end (exit-based); trades entered in training but exiting after the boundary are **purged** to prevent test-period prices leaking into training outcomes.
- **Boundary-crossing test trades** (entered in the test year, exiting after it) are carried to their natural exit using the parameters frozen at entry, and counted in the fold by entry date. Their exit-based classification is also reported as a secondary view.
- **Continuity:** the primary result is a *stitched OOS equity curve* with capital carried across folds; each fold’s parameters apply to new entries only; open positions retain the parameter fingerprint of their entry. Per-fold standalone statistics are secondary.
- No parameter, grid, or rule may be altered after viewing any test fold. A fold’s test period is unread until its parameters are locked and hashed.

13.4 Terminology

Term	Definition
In-sample	Data available for research and parameter selection, including any period whose results influenced design.
Out-of-sample	Data not used to select parameters; strictly the walk-forward test folds and holdout H. A full-period backtest is MUST NOT be called walk-forward OOS.
Forward observation	Data occurring after the framework and configuration were frozen (hash-committed).

Researcher-knowledge caveat. The author has ambient knowledge of Indian market history after 2018 and prior TPQSE research on the same period. Even correct walk-forward mechanics cannot remove this soft leakage; only forward observation can.

14 Robustness Testing

Required battery (all reported regardless of outcome): parameter sensitivity heatmaps (B–F) with neighbour-plateau statistics; ablation matrix ABL; cost sensitivity $1 \times / 2 \times / 3 \times$; capacity/liquidity stress (X4); time-stop and gap-filter stress (X2, X3); adjustment-method stress (X5); Monte Carlo; random control N1/N2; sub-period stability (yearly expectancy); leave-one-year-out expectancy; concentration diagnostics (top-5 trades' share of net P&L; top-sector share; top-symbol share); stability without CA_SUSPECT trades. Robustness is judged by *consistency* across these, never by a single favourable cell.

15 Survivorship Bias

Status for CURRENT_ACTIVE_UNIVERSE: UNRESOLVED. Historical tests include only stocks active today; failed, delisted, suspended, and merged names are absent. Direction and magnitude of the bias cannot be quantified from available data.

Minimum requirement (mandatory). A survivorship audit report per experiment: universe definition, point-in-time flag, count of symbols with data starting after the research start (proxy for IPO/inclusion effects), count of symbols with data ending before the dataset end, and the explicit label SURVIVORSHIP: UNRESOLVED.

Preferred requirement. A point-in-time $Universe_t$ from dated index constituent files or exchange listing/delisting records.

Prohibited. Fabricating delisted companies; synthetic returns for missing names; hiding the limitation; estimating bias magnitude without data. All reports **MUST** carry the survivorship label in the header. Forward observation (Section 18) records the universe snapshot as of each date and therefore avoids survivorship for forward trades.

16 Market Regime Analysis

Deterministic regime classification (frozen before results). Benchmark: NIFTY 50 index daily close B_t (research-only data). With SMA_{200} :

- **Bull:** $B_t > SMA_{200,t}$ and $SMA_{200,t} > SMA_{200,t-20}$.
- **Bear:** $B_t < SMA_{200,t}$ and $SMA_{200,t} < SMA_{200,t-20}$.
- **Sideways:** otherwise.

Regime is evaluated at the signal close t and stored on the trade. Warm-up: 220 index bars. The definition **MUST NOT** change after results are seen.

Market-cap analysis. Classes large/mid/small use AMFI ranking (top 100 / 101–250 / 251+). Store classification source and as-of date. If point-in-time lists are unavailable, current classification is used and every table is labelled CLASSIFICATION_AS_OF_CURRENT; it **MUST NOT** support claims about historical cap segments without that label.

Sector analysis. Per sector: trade count, net P&L, expectancy, win rate, sector-level drawdown. Classification source: NSE industry or vendor sector, recorded with as-of date; current-classification limitation applies.

Minimum sample rule: a regime/sector/cap cell with fewer than 30 trades is reported as LOW_N and excluded from pass/fail criteria.

17 Comparison with TPQSE

Objective: determine whether the two strategies behave differently enough to potentially diversify, *not* which is better.

Measure	Definition
Signal overlap	Jaccard index on (symbol, signal date); also within a ± 3 -session window.
Same-symbol overlap	Fraction of BVEQSE trades in a symbol also held by TPQSE at any time during the hold; and vice versa.
Trade overlap	Fraction of holding-interval days shared per symbol.
Daily return correlation	Pearson and Spearman on daily strategy returns; full-sample, per-year, rolling 126-day.
Trade return correlation	Correlation of R_{net} for overlapping same-symbol trades and for entry-month-aggregated R .
Drawdown correlation	Correlation of underwater curves and fraction of co-drawdown days.
Simultaneous exposure	Distribution of combined invested fraction; days both strategies are at capacity.
Sector concentration	Herfindahl index of combined sector exposure.

Interpretation rules. Both strategies are long-only, share a data vendor and a survivorship-affected universe, and may share market beta, so measured correlation understates common latent biases. A daily return correlation above 0.7 flags LIMITED_DIVERSIFICATION. Results are descriptive; no combined optimization.

17.1 Portfolio combination (post-validation only)

Only if both strategies have independently reached at least “Historically promising”:

Portfolio	Definition
A	TPQSE only.
B	BVEQSE only.
C	50% capital to TPQSE sleeve and 50% to BVEQSE sleeve; sleeves keep independent equity and constraints; no rebalancing in baseline (annual-rebalance variant reported separately).

Allocation weights **MUST NOT** be optimized on the final test period. Combination results **MUST NOT** be reported before both independent validations are complete.

18 Forward Testing

BVEQSE has its own forward engine, state store, and journal, under identifier BVEQSE_v1.0. It **MUST NOT** touch TPQSE state.

Capability	Requirement
Schedule	Run once per session after data is available (default 19:00 IST); ignore runs before the day’s bar is final; runs are idempotent per (version, session date).
Deterministic IDs	<code>signal_id</code> as in Section 6; <code>trade_id</code> = SHA-256(signal_id entry date), 16 hex chars.

Capability	Requirement
Duplicate protection	Unique keys on IDs; reruns update nothing already committed.
State persistence	SQLite (WAL) with transactional writes; state includes cash, positions, pending orders, signals, trades, config hash.
Restart recovery	On start, detect last completed session; process missing sessions in strict order using only data $\leq$ each session; never skip sessions; crash mid-session rolls back.
Universe snapshots	Store the universe (list plus hash) per session or upon change.
Data freshness	Last bar date must equal the last NSE trading session; otherwise flag <code>STALE_DATA</code> and defer processing.
Data revision log	Hash the last 10 sessions of each symbol per run; log any revision of previously stored bars; a signal made on since-revised data is flagged, never silently altered.
Timestamp validation	Reject bars dated after the as-of session; log computation wall-clock time; signals must be computed after close.
Look-ahead checks	Run T-LA1/T-LA2 style truncation check on each run's new signals.
Accounting	Same portfolio, cost, slippage code path as backtest; open-position tracking with frozen stop/target/ATR.
Paper-fill semantics	Pending order persisted at t close; filled using the official $t + 1$ open once that bar is available; exits determined from daily bars with the same stop-first rules.
Immutability	Startup verifies the configuration hash against the frozen release hash and aborts on mismatch.

19 Monitoring and Failure Conditions

Principle: failure conditions *flag* the strategy for human review. They **MUST NOT** modify parameters, pause entries, resize risk, or otherwise alter behaviour.

Condition	Definition (frozen at forward start)	Notes
Expectancy failure	After ≥ 50 completed forward trades, $\overline{R_{net}} < 0$ (FLAG); if the 95% cluster-bootstrap upper bound < 0 (SEVERE).	Evaluated at every milestone.
Slippage failure	Rolling median realized round-trip slippage over the latest 25 trades $> 0.15\%$ ($1.5 \times$ the 0.10% assumption), or $\geq 80\%$ of those trades exceed 0.15% .	Paper fills are modelled, not observed. Evaluable only against an independent reference execution price (e.g., intraday VWAP proxy); otherwise reported NOT_EVALUABLE .
Drawdown failure	Forward drawdown exceeds the 95th percentile of the Monte Carlo max-drawdown distribution (scaled to the same trade count), or an absolute 20% hard threshold.	Both thresholds frozen in config.
Data failure	$> 5\%$ of expected sessions missing over a rolling 60-session window, or $> 5\%$ of signals produced later than scheduled.	Missing-session counts per symbol and aggregate.
Integrity failure	Config hash mismatch, duplicate ID, look-ahead check fail, or negative cash.	Always SEVERE ; forward record marked compromised.

19.1 Milestones

Milestone (completed trades)	Report contents
25, 50, 100, 150, 200	Performance; expectancy with CI; drawdown; data quality; execution quality; slippage; comparison versus historical OOS distribution (percentile of forward metrics within Monte Carlo band); failure-condition status; open positions; parameter/version/config-hash integrity; TPQSE comparison update.

Milestone reports are generated automatically by `journal_generator.py`, are append-only, and are stored with a hash.

19.2 Dashboard (read-only)

A separate process opens the state database in read-only mode and shows: current signals, open positions, historical and forward performance, drawdown, health flags, milestones, data freshness, slippage, failure conditions, TPQSE comparison. It **MUST NOT** import mutating engine code, expose write endpoints, or alter state files.

20 Software Architecture

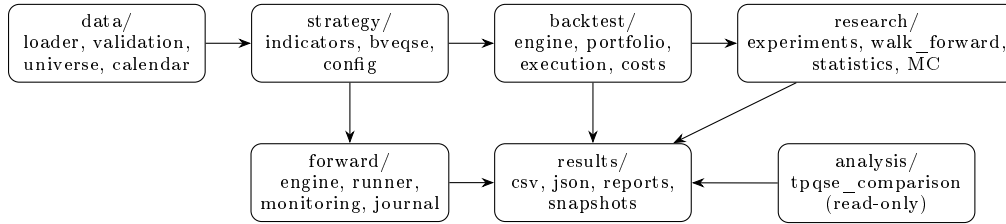


Figure 3: Module dependency overview. Execution, portfolio, and cost code is shared by backtest and forward paths.

Proposed structure (improvements over the sketch: shared core types, separate calendar and corporate-action modules, explicit random-control and ablation modules, read-only TPQSE comparison, state store, dashboard and ML export isolation).

```

bveqse/
core/ types.py (Signal, Order, Fill, Trade), hashing.py, ids.py, seeds.py
data/ loader.py, validation.py, universe.py, calendar.py, corporate_actions.py, symbol_map.py
strategy/ indicators.py, bveqse.py, config.py (frozen, hashed)
backtest/ engine.py, portfolio.py, execution.py, costs.py
research/ experiments.py, ablation.py, sensitivity.py, walk_forward.py,
          statistics.py, monte_carlo.py, random_control.py, regime_analysis.py
analysis/ tpqse_comparison.py (read-only inputs), portfolio_combination.py
forward/ engine.py, runner.py, state_store.py, monitoring.py, journal_generator.py
reporting/ milestone_report.py, survivorship_audit.py, data_quality_report.py
dashboard/ read-only app
ml_export/ dataset builder (no ML in v1.0)
results/ (git-ignored, schema-versioned)
tests/ unit/, integration/, property/, golden/
README.md
  
```

Engineering requirements. Python 3.11+ with pinned dependency versions (lock file); typed, frozen configuration objects (dataclass or pydantic); float64 arithmetic; no global mutable state; the same `execution.py` and `costs.py` used by backtest and forward; deterministic ordering (sort by date then symbol); logging with run IDs.

21 Testing Strategy

Target: **no known test failures before forward deployment.** A pytest suite runs in CI on every change; the forward release is blocked on a fully green suite and a passing golden-dataset regression.

ID	Area	Required assertions
T-IND1	EMA	Matches an independent reference from the formula (SMA seed) to 10^{-9} ; undefined before bar N_E ; reset after missing bar.
T-IND2	ATR/NATR	Wilder recursion, TR uses prior close, seed at bar $N_A + 1$; matches reference.
T-IND3	Volume average	Excludes current bar; NaN invalidates; zero counted.
T-CON	Consolidation	Window excludes bar t ; hand-built cases at threshold equality.
T-CMP	Compression	Bar t excluded from both sides; median window length L_a .

ID	Area	Required assertions
T-BRK	Breakout	Close equal to resistance is not a breakout; intraday high alone is not.
T-SIG	Signal generation	All condition combinations on synthetic data; warm-up boundaries at $W - 1, W$.
T-ENT	Entry	Next-session open; gap-up; missing data; holiday; stale; invalid OHLC; circuit lock.
T-EXI	Stop/target	Stop touch, target touch, stop-first ambiguity, gap-through-stop, gap-through-target execute at open.
T-FRZ	Frozen ATR	S, T, R constant for all holding days.
T-SIZ	Sizing	Uses mark-to-market equity, floors, caps, insufficient cash, $q < 1$ rejection.
T-COST	Costs	Each component per leg vs hand-computed; multipliers; no costs in strategy code.
T-SLP	Slippage	Applied to fill price direction-correctly; round trip $\approx 0.10\%$.
T-PORT	Accounting	Equity identity each day; cash never negative; conservation of P&L.
T-DUP	Duplicate protection	Same session re-run yields identical state and IDs.
T-RST	Restart recovery	Kill mid-run, restart, compare with uninterrupted run byte-for-byte.
T-TS	Timestamp integrity	Future-dated bar rejected; pre-close signal rejected.
T-LA1/2	Look-ahead	Truncation invariance; future-scramble invariance (property-based).
T-WF	Walk-forward boundaries	Purge of straddling training trades; entry-based OOS assignment; no test data in selection; stitched capital continuity.
T-MS	Milestone	Counts, expectancy, CI reproducible from trade log.
T-MC	Statistics	Seeded bootstrap/MC reproducible; p-value formula on known synthetic nulls.
T-EQ	Vector vs event-driven	Equivalent signals and trades on sampled data.
T-GOLD	Golden regression	Frozen small dataset with expected trades/metrics; any change requires explicit re-baselining review.

Pre-implementation tests to specify first (Appendix A.7): synthetic price generators with known breakout/no-breakout structure; hand-computed indicator vectors; execution scenario tables.

22 Reproducibility

Every experiment stores a `run_manifest.json`: experiment ID, strategy version, full configuration, configuration hash, data source, data pull timestamp and per-symbol data hashes, universe version and snapshot hash, random seeds, Python and library versions, git commit, and cost-config version.

Configuration hash: SHA-256 of canonical JSON (sorted keys, normalized floats, UTF-8) covering strategy, execution, costs, portfolio constraints, universe reference, and data specification.

Experiment ID: `BVEQSE_v1.0__<EXP>__<hash8>__<yyyymmdd>`.

Seeds: a master seed (default 20260929) is recorded; each stochastic procedure uses `numpy.random.default_rng` seeded from SHA-256(master seed || experiment ID || procedure name). Re-running with identical inputs **MUST** reproduce all outputs exactly.

22.1 Version control and immutability

Change	Version rule
Documentation or reporting only, no result change	Same version, changelog entry.
Bug fix altering historical results	Errata record plus rerun; if forward trades are affected, new minor version and a new forward record.

Change	Version rule
Minor rule change (e.g., parameter, filter, cost model)	BVEQSE_v1.1; new forward record.
Change to signal, execution, or risk architecture	BVEQSE_v2.0.

Once forward testing begins, v1.0 is immutable: source tag, config hash, and dependency lock are committed and verified at startup. Silent modification is a protocol violation.

23 Output Specifications

```
results/<experiment_id>/
run_manifest.json trades.csv signals.csv portfolio_history.csv metrics.json
walk_forward.csv monte_carlo.csv sensitivity.csv regime_analysis.csv
cost_sensitivity.csv random_control.csv ablation.csv
universe_snapshots/ data_quality/ reports/
```

All CSVs use ISO dates, UTF-8, a schema version header file, and stable column order.

23.1 trades.csv (one row per closed or censored trade)

Column	Type	Description
trade_id, signal_id	str	Deterministic IDs.
symbol	str	Mapped NSE symbol.
signal_date, entry_date, exit_date	date	Signal close, fill, exit dates.
entry_price, exit_price	float	Fill prices after slippage.
open_ref_entry, exit_ref	float	Reference prices before slippage.
atr_signal, stop, target, r_price	float	Frozen risk definition.
quantity	int	Shares.
exit_reason	enum	STOP, TARGET, GAP_STOP, GAP_TARGET, TIME_STOP, CENSORED, DATA_TERMINATION.
gross_pnl, fees, net_pnl	float	INR.
r_frictionless, r_price_mult, r_net	float	R definitions of Sec. 8.
holding_sessions	int	Sessions held.
mfe_r, mae_r	float	In R units.
gap_pct	float	Entry gap vs signal close.
regime, sector, mcap_class	str	Research-only tags.
fold_id, oos_flag, split_label	str	Walk-forward assignment.
flags	str	Pipe-separated, e.g., CA_SUSPECT.
config_hash, strat-egy_version	str	Integrity fields.

23.2 signals.csv, portfolio_history.csv, other files

File	Required columns
signals.csv	signal_id, symbol, signal_date, status, reject_reason, atr_signal, natr, natr_median_ref, range_L, hh_prev, close, volume, vol_avg_prev, vol_ratio, ema50, ema50_lag5, ema_dist_pct, median_traded_value, regime, sector, mcap_class, rank, config_hash.
portfolio_history.csv	date, cash, market_value, equity, open_positions, exposure_pct, daily_return, drawdown, realized_pnl_cum, fees_cum.
metrics.json	Nested keys: trade_level, portfolio_level, distribution, ci (method, n_boot, seed), survivorship_status, n_trials, dsr.
walk_forward.csv	fold, train_start, train_end, test_start, test_end, selected_params, train_score, train_trades, test_trades, test_expectancy_r, test_return, flags.
monte_carlo.csv	sim_id, mode, terminal_equity, max_drawdown, longest_losing_streak; plus summary rows for probabilities.
sensitivity.csv	experiment_id, parameter_vector, trades, expectancy_r, profit_factor, cagr, sharpe, max_dd, neighbour_positive_fraction.
regime_analysis.csv	dimension (regime/sector/mcap), bucket, trades, win_rate, expectancy_r, net_pnl, max_dd, low_n_flag.
cost_sensitivity.csv	multiplier, cagr, sharpe, max_dd, net_pnl, expectancy_r.
universe_snapshots/	YYYYMMDD_symbols.csv plus hash manifest.

24 Future ML Extension

v1.0 contains no machine learning. The trade database is designed to support it later: each setup stores all signal-time features (range, NATR and ratio to median, volume ratio, breakout distance $C_t/HH - 1$, EMA distance and slope, gap, 60-day traded value, distance to 52-week high, day of week, regime, sector, cap class), a **feature_available_at** timestamp, and outcomes (entry, stop, target, MFE, MAE, holding period, exit reason, R_{net}). Rejected and suppressed signals are retained with their reasons, enabling unbiased candidate-set datasets. Future *TPQSE-ML* / *BVEQSE-ML* must be a separate research layer with its own PRD, purged and embargoed time-series validation, and must never modify BVEQSE v1.0 rules or state. An `ml\export` module produces read-only datasets.

25 Acceptance Criteria

A strategy is **not** validated because net P&L is positive, CAGR is high, win rate is high, the backtest looks good, or an ML model has high accuracy. Thresholds below are pre-registered placeholders that **MUST** be frozen in config before any test is run; changing them after results is a protocol violation.

Status	Entry requirement
Research candi- date	PRD frozen; implementation complete; all tests green; baseline Experiment A runs reproducibly; data-quality and survivorship audits produced.
Historically promising	On stitched walk-forward OOS (2021–2025), net of $1\times$ costs and slippage all hold: (1) ≥ 200 OOS trades; (2) net expectancy $\geq 0.10R$ with cluster-bootstrap 95% lower bound > 0 ; (3) profit factor ≥ 1.2 ; (4) positive expectancy in ≥ 4 of 5 test folds; (5) expectancy > 0 at $2\times$ costs; (6) neighbour-plateau: $\geq 60\%$ of one-step neighbours of the baseline (and of the fold-selected cells) have positive net expectancy; (7) random-control $p \leq 0.05$ vs N1 and $p \leq 0.10$ vs N2; (8) Deflated Sharpe probability ≥ 0.90 ; (9) no regime bucket with ≥ 30 trades and expectancy $< -0.15R$; (10) MC 95th-percentile max drawdown $\leq 35\%$ and $P(Eq_{term} < Eq_0) \leq 10\%$; (11) no single symbol above 10% and top-5 trades below 25% of net P&L; (12) holdout H, evaluated once, not contradicting (expectancy not significantly negative). Survivorship remaining UNRESOLVED caps status here and the label is mandatory.
Forward obser- vation	Status “Historically promising”; configuration hash committed and tagged; forward engine tests green; forward start date recorded. TPQSE comparison completed.

Status	Entry requirement
Validated	≥ 100 completed forward trades and ≥ 12 months; forward expectancy > 0 ; forward metrics within the historical MC 5th–95th percentile band; no active SEVERE flags at the last milestone; data-failure and drawdown conditions never triggered (or reviewed and closed); slippage within tolerance where evaluable. Forward trades use point-in-time universes, which addresses survivorship for the forward record only.
Failed	Any historical criterion (2)–(5) or (7) fails definitively; or forward expectancy failure at ≥ 50 trades with upper CI bound < 0 ; or an integrity failure. “Failed” outcomes are retained permanently, not deleted.

“Validated” **MUST NOT** be used unless every listed condition is objectively satisfied and documented.

26 Risks and Limitations

Risk	Description and mitigation
Survivorship (UNRESOLVED)	Current-active universe; direction and size unknown. Mitigated only by labelling and forward observation.
Data vendor limits	Unofficial yfinance; revisions, gaps, imperfect corporate-action adjustments. Mitigated by validation gates and hashes.
Adjusted-price artifacts	Executable price levels, tick sizes, lots and stops in INR are approximate.
Execution optimism	Daily bars cannot capture opening auctions, circuit locks, partial fills, liquidity impact. Mitigated by slippage, stop-first, gap rules, capacity caps.
Multiple testing	Large grid and many experiments inflate false positives. Mitigated by reduced walk-forward grid, plateau selection, n_{trials} , deflated Sharpe.
Statistical dependence	Trades cluster in time and share market shocks; effective sample size is smaller than trade count. Mitigated by cluster bootstrap and block MC.
Small samples per fold	Yearly tests may have few trades; wide intervals expected.
Regime dependence	Long-only breakouts may only work in bull regimes; possible high correlation with TPQSE.
Researcher knowledge leakage	See Section 13.
Cost model drift	Regulatory fees change; placeholders must be verified and versioned.
Operational	Forward runs depend on timely data; failures flagged, never patched silently.
Non-stationarity	Past distributions may not persist; MC and CIs are conditional.

27 Implementation Roadmap

Phase	Deliverables	Exit gate	Depends on
0	Specification freeze; threshold registry; synthetic test data spec	PRD reviewed; grids and criteria hashed	–
1	Data layer: loader, validation, calendar, universe, symbol map	Data-quality and survivorship audits generated	0
2	Indicators, strategy, config hashing	T-IND, T-CON, T-CMP, T-BRK, T-SIG, T-LA green	1
3	Execution, costs, portfolio, backtest engine	T-ENT, T-EXI, T-SIZ, T-COST, T-PORT, T-EQ green; Experiment A reproducible	2

Phase	Deliverables	Exit gate	Depends on
4	Research suite: experiments, ablation, sensitivity, statistics, MC, random control	T-MC green; all experiments run from manifest	3
5	Walk-forward and holdout tooling	T-WF green; frozen selection rule executed once	4
6	TPQSE comparison (read-only)	Comparison report; no TPQSE writes verified	3
7	Forward engine, state store, monitoring, journal	T-DUP, T-RST, T-TS green; dry-run over history equals backtest	3
8	Freeze v1.0: hash, tag, dependency lock, acceptance review	Status “Forward observation” only if criteria met	5, 6, 7
9	Forward operation and milestone reports at 25/50/100/150/200	Status transitions per Section 25	8

A Appendix

A.1 Ambiguity register and resolutions

ID	Ambiguity	Deterministic resolution
A1	ATR_t compression uses the break-out bar’s inflated TR	Compression on $NATR_{t-1}$ vs prior 20; stop uses ATR_t as specified.
A2	Raw versus price-normalized ATR	Normalized (NATR); raw variant not in baseline.
A3	Stop defined from unknown entry	Stop/target computed at fill; $R = kATR_{signal}$.
A4	Which equity for sizing	Close-marked equity of $d - 1$, frozen at start of d .
A5	Do stops apply on entry day	Yes, using entry-day high and low after the open.
A6	No time stop specified	None in baseline; X2 tests variants; end-of-data censoring defined.
A7	Position and capacity limits absent	P13–P16 explicit, frozen.
A8	Signal ranking on capacity conflict	P19 rule with deterministic tie-breaks.
A9	Consolidation and breakout windows both 20	Tied via L in v1.0; interaction with ρ acknowledged.
A10	Missing bar inside window	Window/recursion undefined; no imputation.
A11	Limit-exit slippage	Applied by default; flag configurable.
A12	“Negative ending equity” in MC	Replaced by terminal-loss and ruin-threshold probabilities.
A13	Training meaning	Parameter selection on training trades only; purge straddling trades.
A14	Adjusted versus unadjusted prices	Split/bonus-adjusted, not dividend-adjusted baseline; X5 sensitivity.
A15	Paper-trading slippage observability	Modelled; slippage failure often NOT_EVALUABLE.

A.2 Look-ahead bias sources

Resistance using bar t ’s own high; ATR or NATR window including bar t where prior state is intended; volume average including V_t ; trend evaluated with revised or adjusted-later data; percentile or normalization over the full sample; regime, sector, cap classification from the future; universe membership from today; next-day open used for signal filters; end-of-day equity used for same-day entries; rounding or vendor back-adjustment revisions between runs; parameter selection viewing test folds; researcher knowledge of post-2018 outcomes.

A.3 Survivorship sources

Current-active universe; absent delisted, suspended, merged, renamed symbols; index-inclusion timing; IPO recency; vendor dropping dead tickers; symbol reuse.

A.4 Backtesting error register

Filling at signal close; ignoring gaps; optimistic intrabar order; slippage as P&L subtraction; costs omitted on one leg; cash-based sizing; leveraged negative cash; ATR recalculated after entry; double counting simultaneous fills; carrying stale data forward; wrong calendar; adjusted prices treated as executable; deleting rejected signals; fold leakage through straddling trades.

A.5 Statistical pitfalls

Multiple testing and data snooping; overlapping and clustered trades; autocorrelation; heavy tails; non-stationarity; small folds; reporting best cell; Sharpe ignoring time-in-market; bootstrap assuming independence; treating random-control pass as proof; confusing in-sample and OOS; survivorship inflating win rates.

A.6 External dependencies

yfinance (unofficial, rate-limited, revisable); NSE trading calendar; NSE index constituents (historical availability limited); AMFI market-cap lists; sector classification source; broker/exchange/SEBI fee schedules; Python scientific stack (pandas, numpy, scipy, pytest, hypothesis, SQLite); NIFTY 50 index data; read-only TPQSE artifacts.

A.7 Configurable versus frozen

Class	Items
Configurable (re-search)	Grid values within bounds (Sec. 5), cost multipliers, portfolio constraints for stress runs, universe provider, starting equity, price series choice.
Frozen at PRD freeze	Baseline rules C1–C8; execution order; stop-first rule; frozen-ATR rule; sizing definition; regime definition; walk-forward folds and selection rule; acceptance thresholds; failure thresholds; seeds derivation; random-control definition; output schemas.
Frozen at forward start	Entire BVEQSE_v1.0 config and hash, dependency lock, universe policy, cost config with effective date.

A.8 Example configuration skeleton (illustrative, not code)

```
{
  "strategy_version": "BVEQSE_v1.0",
  "signal": {"ema_period": 50, "ema_slope_lag": 5, "lookback": 20, "range_threshold": 0.10,
    "atr_period": 14, "compression": {"mode": "median", "window": 20},
    "volume": {"multiple": 1.5, "window": 20}, "liquidity_floor_inr": 100000000},
  "risk": {"stop_atr_multiple": 1.5, "target_r": 2.0, "risk_fraction": 0.005,
    "max_positions": 10, "max_weight": 0.20, "adv_participation": 0.02},
  "execution": {"slippage_per_leg": 0.0005, "slippage_on_limit_exits": true,
    "intrabar_ambiguity": "STOP_FIRST", "circuit_lock_heuristic": true},
  "data": {"source": "yfinance", "auto_adjust": false, "series": "split_bonus_adjusted",
    "history_start": "2017-01-01"},
  "universe": {"provider": "CURRENT_ACTIVE_UNIVERSE", "point_in_time": false},
  "reproducibility": {"master_seed": 20260929, "n_bootstrap": 10000, "n_monte_carlo": 10000}
}
```

A.9 Glossary

R: initial risk per share. **MFE/MAE**: maximum favourable/adverse excursion. **OOS**: out-of-sample. **PIT**: point-in-time. **ADV**: average daily volume. **DSR**: deflated Sharpe ratio. **Censored trade**: still open at data end.